

NEURAL BELLMAN OPERATORS

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We study neural policy evaluation and improvement in stochastic economic models with endogenous preferences, recursive utility, and strategic interaction. Independent verification separates the implemented policy from its critic and from any corrective action search. For stopped diffusions, one-sided smooth witnesses bound continuous policy regret without requiring a smooth optimal value at a controlled exit boundary. We establish error-calibrated comparative statics for costly preference adjustment and a complete constrained Hamiltonian selection rule for the economic application. Executed interval certificates cover retained policies in the original continuous economy, although the resulting enclosures remain too wide for precise welfare comparisons. A learned-actor experiment on the same state, control, covariance, and exit geometry attains continuous regret bounds below .00113 for an explicitly modified payoff. Thresholded finite correction preserves substantially more of a retained long-horizon neural policy while accounting for the full action-search cost. In eight- and sixteen-dimensional coupled nonconvex control problems, paired experiments favor NBO over a pinned author implementation of SOC-MartNet at the declared short training budget. The theoretical scope, numerical precision, and computational comparisons are stated separately and tied to complete reproducible records.

KEYWORDS: Stochastic control, Neural policy iteration, Endogenous preferences, Recursive utility, Verification, Computational economics.

1. INTRODUCTION

How accurately must an economic decision rule be computed before a welfare comparison is informative? This question is especially consequential when the decision maker changes not only consumption and investment but also a preference state. A small error in a fitted value function need not imply a small policy loss, and a numerically stable policy surface need not distinguish an adjustment-cost effect from an approximation error. The difficulty is compounded when an economic boundary is an actual termination event rather than an artificial edge of a training domain.

This paper develops Neural Bellman Operators (NBO) as a model-based framework for separating policy evaluation, feasible policy improvement, and verification of the implemented decision rule. The economic application combines stochastic risk aversion, consumption, portfolio choice, and costly preference adjustment. Its stopping and settlement contract is specified as part of the economy. The mathematical results identify three distinct objects: an exact policy-improvement sequence, its neural implementation, and an independently evaluated policy. None is substituted for another in the error bounds.

Our first result provides a continuous-time verification pair. A supersolution controls the value of every admissible policy, whereas a subsolution controls the payoff of the particular policy to be implemented. Their difference, together with explicitly signed residual and boundary terms, bounds regret. The argument uses smooth test functions, not smoothness of the optimal value. It therefore does not require continuity of the optimal value at an irregular controlled exit boundary. We give both a direct continuous-generator implementation and an operator-enclosure decomposition for a discretized implementation. The latter separates time, action, state interpolation, quadrature, exit, inner-solve, and arithmetic errors; a refinement difference is not one of these bounds.

The second result connects computational accuracy to an economic comparative static. Raising the price of preference adjustment lowers the optimal discounted adjustment budget. For policies that are only approximately optimal, the possible reversal in this ordering is bounded by the sum of their regret bounds divided by the change in price. Value enclosures also bound optimal adjustment budgets through secants and bound the welfare gain from access to the adjustment technology. These statements make the required numerical precision depend on the economic comparison, rather than on a common uncalibrated stopping tolerance.

The computational analysis has complementary components. We re-evaluate the retained neural NDU policies, price the full finite-action safeguard, and separate time, state, and action refinements. We construct outward-interval verification pairs for their deployment in the original continuous diffusion. Those pairs are valid but too conservative to resolve the stated economic precision target. A second, explicitly modified-payoff experiment trains a bounded actor on exactly the same state, control, covariance, and exit geometry and obtains an all-state continuous regret certificate. Finally, a coupled nonconvex control problem in eight and sixteen dimensions is compared at a fixed CPU budget with the pinned author implementation of SOC-MartNet. The comparison uses actual neural controls, no analytical greedy-action labels, paired independent policy rollouts, and a declared six-seed design. Its outcome measures realized cost at that budget, not distance to an unknown optimum.

Policy iteration, neural PDE evaluation, and their combination have important antecedents. Jacka and Mijatović (2015) study continuous-time policy improvement. Kim, Cho, Kim, and Kim (2026) alternate soft evaluation and improvement for entropy-regularized stochastic control and separate iteration, policy-network, and residual errors. The updated interior analysis of Kim, Kim, Kim, and Cho (2026) treats stationary second-order equations with bounded-domain training and unresolved outer information; its interior bounds and its discussion of collocation–closed-loop mismatch are directly relevant. Lee and Kim (2025) combine policy iteration with operator learning to reuse a representation across terminal data. Cai, Fang, and Zhou (2025) train stochastic control and value networks through martingale conditions without explicitly computing the Hamiltonian infimum. NBO does not claim priority for neural policy iteration or for residual verification. Its contribution here is the combination of stopped economic-model verification, explicitly costed feasible correction, and error-calibrated preference-adjustment comparisons. Section 7 gives the comparison at the level of assumptions and implemented experiments.

Recursive utility, sophisticated temporal selves, and dynamic games remain part of the framework. They do not share an identical evaluation equation or an identical optimality notion. We retain their aggregators, equilibrium conventions, proofs, and computational records, and state precisely how a gain bound changes when the evaluation operator is nonlinear or when rivals are frozen. The supplementary material preserves the complete preceding R4 manuscript and technical derivations; the current manuscript integrates the R5 evidence and the new continuous verification and external-comparison results.

2. AN ECONOMY WITH COSTLY PREFERENCE ADJUSTMENT

The state is $s = (u, x)$, where u is relative risk aversion and x is financial wealth. An admissible control is a progressively measurable process $a = (c, \theta, p)$ taking values in

$$A = [.05, .8] \times [-.2, .2] \times [-.5, .8].$$

Before termination the state satisfies

$$du_t = \theta_t dt + .05 dZ_t^u, \tag{1}$$

$$dx_t = [(0.02 + 0.06p_t)x_t - c_t]dt + 0.2p_tx_t dZ_t^x, \quad d\langle Z^u, Z^x \rangle_t = -0.25 dt. \quad (2)$$

The horizon is $T = 1$, the open state domain is $D = (1.2, 2.8) \times (.5, 2)$, and the discount rate is $\rho = .04$. The process is stopped at $\tau = T \wedge \inf\{r \geq t : S_r \notin D\}$. Its running payoff and settlement are

$$\ell_k(u, c, \theta) = \frac{c^{1-u}}{1-u} - \frac{k}{2}\theta^2, \quad k > 0, \quad (3)$$

$$G(u, x) = -0.02(u - 2)^2 + .1 \log x, \quad g(t, u, x) = G(u, x) - 8(1 - t). \quad (4)$$

Thus

$$J_k^\pi(t, s) = \mathbb{E}_{t,s}^\pi \left[\int_t^\tau e^{-\rho(r-t)} \ell_k(S_r, a_r) dr + e^{-\rho(\tau-t)} g(\tau, S_\tau) \right], \quad V_k(t, s) = \sup_\pi J_k^\pi(t, s). \quad (5)$$

The feasible controls, state dynamics, stopping rule, and settlement do not depend on k . Bounded controls and locally Lipschitz state coefficients give a well-defined stopped state equation for the feedbacks used below. Bounded payoffs on the stated domain give finite values. The cardinal normalization and consumption units are held fixed across comparisons. A transformation of utility depending on the controlled state u is generally a different economy.

The early termination fee is a specified liquidation covenant, not an estimated preference parameter. At termination there is no subsequent wealth injection or continuation consumption. This distinction is essential: a constant preference diffusion cannot be confined to the stated rectangle by merely clipping simulated states. At the lower wealth boundary the largest admissible drift is -0.016 , so that the original consumption floor cannot support a silently viable self-financing rectangle. We do not replace this economy by reflection.

At the upper wealth boundary, the admissible choice $p = 0$ has zero wealth diffusion and inward drift $.04 - c < 0$. Consequently, a theorem that assumes boundary regularity uniformly over all controls cannot simply be applied to this model. We neither identify the interior trace of V_k with the prescribed immediate-settlement value without proof nor assume a globally smooth optimal value. The verification pair in Section 4 uses inequalities on the actual stopped trace and remains meaningful in this setting.

For a smooth test function v , define

$$\mathcal{L}^a v = \theta v_u + [(0.02 + 0.06p)x - c]v_x + .00125v_{uu} + .02p^2x^2v_{xx} - .0025pxv_{ux}, \quad (6)$$

$$R_a v = v_t + \mathcal{L}^a v + \ell_k - \rho v. \quad (7)$$

The interior HJB equation is $\sup_{a \in A} R_a V_k = 0$ in the appropriate solution sense. All covariance terms, including the mixed derivative, are retained. A negative shock correlation by itself does not establish the sign of the optimal portfolio response, and dividing a re-solved V_u by k does not establish a pointwise ordering of adjustment policies.

PROPOSITION 1—Complete control selection for a fixed jet: *Let v have the derivatives in (7). A global maximizer of $R_a v$ is obtained separately in the three control components. If $v_x > 0$, $c^* = \text{proj}_{[.05, .8]}(v_x^{-1/u})$; otherwise $c^* = .8$. Also $\theta^* = \text{proj}_{[-.2, .2]}(v_u/k)$. Write*

$$h_1 = .06xv_x - .0025xv_{ux}, \quad h_2 = .04x^2v_{xx}.$$

If $h_2 < 0$, then $p^ = \text{proj}_{[-.5, .8]}(-h_1/h_2)$. If $h_2 \geq 0$, choose an endpoint maximizing $h_1 p + h_2 p^2/2$, with a fixed tie rule.*

This selection rule includes nonpositive marginal wealth value, zero curvature, convex portfolio objectives, and constrained optima. It removes action discretization from a continuous-generator audit for this particular model. It is not a closed-form optimizer of the killed-Euler, interpolated, finite-step Bellman target, and it is not a global convergence theorem for shared neural parameters. The underlying derivative rule was present in the retained derivations; the present implementation tests every curvature branch explicitly.

3. EVALUATION, IMPROVEMENT, AND FINITE CORRECTION

For a fixed policy π , policy evaluation solves $R_\pi v = 0$ with the economic stopping contract. Improvement chooses a feasible control maximizing the Hamiltonian of that fixed evaluation. The exact operators are denoted $\mathcal{E}$ and $\mathcal{I}$, and exact policy iteration is $\pi_{j+1} = \mathcal{I}(\mathcal{E}\pi_j)$. If a classical evaluation exists, $R_{\pi_{j+1}}(\mathcal{E}\pi_j) \geq 0$. Verification then gives $J^{\pi_{j+1}} \geq J^{\pi_j}$.

An exact limiting argument requires more than this monotonicity. Under compactness of the policy class, continuity of evaluation into a precompact $C^{1,2}$ jet class, closure under improvement, continuity of the chosen selector at cluster points, and comparison for the optimality equation, each cluster-point value is optimal. These are hypotheses on the exact operators. They do not follow from successful stochastic-gradient training. Appendix B gives the limiting argument and retains the distinctions between parameter stationarity, policy convergence, and value convergence.

3.1. The raw neural algorithm

A critic v_ξ is trained against fixed-policy targets. In a continuous-generator implementation these are $R_\pi v_\xi$; in a discrete implementation they are the appropriate one-step conditional expectations. The actor improves a frozen critic. Freezing the critic's parameters does not mean deleting the derivative of its output with respect to the controlled next state. Conversely, no actor update may change the critic parameters or a rival's policy parameters through the improvement objective. Independent policy evaluation is a third operation and supplies neither optimal-action nor optimal-value training labels.

In the NDU implementation inherited from R5, the raw actor has three bounded outputs and two tanh hidden layers. It is trained without enumerating actions, but its training states lie on a lattice. It is therefore a continuous-action, lattice-trained implementation, not a mesh-free experiment. The high-dimensional experiment below instead uses sampled states and Gaussian successor samples. The two implementations share the separation of evaluation and improvement; their verification costs are different.

3.2. A certificate for the declared finite economy

Let $\mathcal{T}_n^a w = r_n^a + \delta P_n^a w$, where P_n^a is a positive substochastic continuation matrix and $\delta = e^{-\rho\Delta}$. Termination payoffs are included in r_n^a . Let $\hat{\pi}$ be a feasible policy and v_n a critic, with terminal discrepancy at most b . Define

$$e_n = \|v_n - \mathcal{T}_n^{\hat{\pi}} v_{n+1}\|_\infty, \quad \eta_n = \|\max_a \mathcal{T}_n^a v_{n+1} - \mathcal{T}_n^{\hat{\pi}} v_{n+1}\|_\infty.$$

THEOREM 1—Finite gain accounting: *On this finite economy,*

$$\|V_{h,n}^* - V_{h,n}^{\hat{\pi}}\|_\infty \leq 2\delta^{N-n}b + \sum_{j=n}^{N-1} \delta^{j-n}(2e_j + \eta_j). \quad (8)$$

If $v = V_h^{\hat{\pi}}$ is independently evaluated with the exact finite terminal data, $e_j = b = 0$. A state-dependent bound is obtained by setting $B_N = 0$ and

$$B_n(s) = g_n(s) + \delta \max_a (P_n^a B_{n+1})(s), \quad g_n = \max_a \mathcal{T}_n^a V_{h,n+1}^{\hat{\pi}} - V_{h,n}^{\hat{\pi}}.$$

Then $0 \leq V_{h,n}^* - V_{h,n}^{\hat{\pi}} \leq B_n$.

Here the maximum must range over the entire declared candidate economy. For the retained NDU audit this is the 125-point action grid together with the frozen continuous neural proposal at every state and date. The certificate does not cover all continuous controls merely because one candidate originated in a neural network. Floating-point computation of the finite envelope is recorded separately from the real-arithmetic statement of the theorem.

3.3. The safeguarded algorithm and its cost

The safeguarded algorithm first evaluates the raw policy, computes all candidate gains, and corrects states with gain above a specified threshold. It re-evaluates the corrected policy and stops only when its reported envelope meets the target. In the new re-audit the threshold is

$$\gamma = \frac{\varepsilon}{N-1} \cdot 2 \sum_{n=0}^{\infty} \delta^n. \quad (9)$$

The factor two leaves a margin relative to the sufficient condition $\max_{n,s} g_n(s) \leq \gamma$. The actual stopping test remains the recomputed envelope, not an optimal reference value. A stopped run that does not meet that test is a failed precision test.

PROPOSITION 2—When corrections can become infrequent: *For M audited state-date decisions with nonnegative gains g_m , the share whose gain exceeds $\gamma > 0$ is at most $(M\gamma)^{-1} \sum_m g_m$. Thus a vanishing mean gain is sufficient for a vanishing correction share at a fixed threshold. For thresholds $\gamma_j \downarrow 0$, the sufficient condition is $M_j^{-1} \sum_m g_{j,m} = o(\gamma_j)$.*

This is a conditional mechanism, not evidence that longer training has already made the retained corrections negligible. It distinguishes a useful limit statement from an unmeasured training-dose claim. Every audit still enumerates the candidate actions. With N dates, S lattice states, and A grid actions, the implementation uses $NS(A+2)$ value-action evaluations per audit, including policy evaluation and the extra proposal, and $NS(A+1)$ envelope-action evaluations. Each NDU action uses four shock branches and bilinear interpolation. There is no dimension-free cost claim for this safeguard.

4. VERIFICATION OF THE CONTINUOUS ECONOMY

The object implemented in continuous time must be stated before it can be certified. For the retained NDU proposals, we use their feasible bilinear interpolation on the exact uniform state grid. On each decision interval the feedback function is frozen in time, but is evaluated at the current state continuously: $a_r = \pi_n(S_r)$. This is not a simulation that holds the initial action until the next date. It defines a bounded, piecewise-time, Lipschitz state-feedback rule for the original Brownian dynamics and exact first-exit contract.

Write $C_\rho(h) = (1 - e^{-\rho h})/\rho$, with $C_0(h) = h$.

ASSUMPTION 1—Admissible verification witnesses: U and L are bounded, continuous up to the stopped trace, and $C^{1,2}$ in the interior of each time slab. Their derivatives permit localized Itô verification and the stopped expectations are uniformly integrable. The policy π has a well-defined stopped state equation. At interior slab interfaces either the witnesses are continuous or they satisfy $U(t^-, s) \geq U(t^+, s)$ and $L(t^-, s) \leq L(t^+, s)$.

THEOREM 2—A one-sided continuous verification pair: Under Assumption 1, suppose throughout the open domain

$$\sup_{a \in A} R_a U \leq \epsilon_U, \quad R_\pi L \geq -\epsilon_L,$$

and on the actual terminal and exit trace $U \geq g - b_U$ and $L \leq g + b_L$, where all four constants are nonnegative. Then

$$V(t, s) \leq U(t, s) + b_U + C_\rho(T - t)\epsilon_U, \quad (10)$$

$$J^\pi(t, s) \geq L(t, s) - b_L - C_\rho(T - t)\epsilon_L, \quad (11)$$

$$0 \leq V - J^\pi \leq U - L + b_U + b_L + C_\rho(T - t)(\epsilon_U + \epsilon_L). \quad (12)$$

The true value need not belong to $C^{1,2}$ or be continuous at every controlled boundary point.

The two witnesses need not be neural networks, and need not have the same approximation error. A trained network can propose either witness, but verification requires an enclosure over the entire stated domain and the full control set. Passing a random test set is not such an enclosure. The symmetric residual certificate is recovered by taking $U = L = v$, with $|R_\pi v| \leq e$, $\sup_a R_a v - R_\pi v \leq \eta$, and $|v - g| \leq b$: regret is at most $2b + C_\rho(2e + \eta)$.

Spatially localized errors can be handled without replacing them by a sampled mean. If $\sup_a R_a U \leq q_U(s, t)$, choose a nonnegative potential z_U with zero-or-positive terminal/exit trace such that $z_{U,t} + \mathcal{L}^a z_U - \rho z_U \leq -q_U$ for every action. Then $U + z_U$ is an exact upper witness. The analogous potential for the fixed policy converts L into a lower witness. These occupation-sensitive potentials identify a constructive route to tighten conservative global certificates; they themselves must be verified rather than estimated on favorable trajectories.

4.1. An explicit original-payoff construction

For G in (4), Proposition 1 gives $c = .8$, $p = .8$, and $\theta = -.02(u - 2)$ when $k = 2$. A 160-by-160 outward-interval partition bounds $\sup_{s,a} R_a G$ above by $-\alpha$, with $\alpha > 0$. For each frozen policy we likewise bound $R_\pi g$ below by $-\beta$ on every state cell and time slab. Consequently,

$$U = G - \alpha C_\rho(1 - t), \quad L = g - \beta C_\rho(1 - t) \quad (13)$$

are valid exact-sign witnesses. Since $\alpha < 8$, $U \geq g$ on the stopped trace, while $L \leq g$. Their date-zero difference is

$$8 + (\beta - \alpha)C_\rho(1). \quad (14)$$

The interval program encloses logarithms, powers, products, and all bilinear-action component ranges. Input network outputs are frozen binary64 constants interpreted as exact real numbers; the state grid is the exact uniform rational grid. The returned bound is rounded outward again when converted to binary64. This certifies the specified real-arithmetic feedback, not every floating-point implementation of the stochastic differential equation.

TABLE I
ORIGINAL-PAYOFF CONTINUOUS NDU WITNESS BOUNDS

N	Interval cells	α	β	Regret upper bound
6	4320	0.864832	7.078878	14.091405
12	8640	0.864832	1.211230	8.339561

Retained raw seed-10 nodal policies, deployed as continuously reevaluated bilinear feedback on each time slab. Bounds cover the original continuous Brownian model and exact first exit using outward interval arithmetic. Both bounds exceed the .01 precision target.

The original-payoff bounds in Table I are numerically wide. They replace an absent continuous certificate by an executed, valid certificate, but do not establish regret below .01 or .05. In particular they are not a substitute for a sufficiently narrow welfare enclosure. The theorem and its implementation make that distinction testable rather than hiding it in a missing discretization term.

4.2. A complete discretization accounting route

There is a second route to continuous verification. Let $\mathcal{B}_n^*$ be the exact continuous-time optimal Bellman semigroup over one decision interval, allowing all admissible within-interval controls, and let $\mathcal{B}_n^\pi$ be the exact semigroup for the declared continuously implemented policy. Lift the finite optimal and finite policy values to bounded functions w_n^* and w_n^π by a positive norm-one extension. Suppose certified enclosures give

$$\|w_n^* - \mathcal{B}_n^* w_{n+1}^*\|_\infty \leq \kappa_n^*, \quad \|w_n^\pi - \mathcal{B}_n^\pi w_{n+1}^\pi\|_\infty \leq \kappa_n^\pi,$$

with terminal discrepancies b^* and b^π . A telescoping semigroup argument yields

$$\|V_0 - J_0^\pi\|_\infty \leq B_h + \delta^N(b^* + b^\pi) + \sum_{n=0}^{N-1} \delta^n(\kappa_n^* + \kappa_n^\pi), \quad (15)$$

where B_h bounds the lifted finite optimal-policy gap. Each κ can be enclosed by a declared chain of intermediate operators separating time freezing (including within-step feedback), restriction of actions, state interpolation, shock quadrature, first exit and settlement timing, inner optimization, and roundoff. The sum is valid when each adjacent operator difference is bounded on the same lifted continuation function. A null term is unknown, not zero. A coarse-fine solution difference alone bounds none of the uncomputed tails. Nor does a regular-domain scheme theorem automatically apply at an irregular controlled exit boundary. The direct generator pair (12) bypasses these scheme errors by auditing the continuous implemented policy itself; it does not assert that the errors of the training scheme vanish.

5. ECONOMIC COMPARISONS AT FINITE NUMERICAL ACCURACY

Fix an initial state and suppress it from the notation. For each feasible policy write

$$A(\pi) = \mathbb{E} \left[\int_t^\tau e^{-\rho(r-t)} \frac{c_r^{1-u_r}}{1-u_r} dr + e^{-\rho(\tau-t)} g(\tau, S_\tau) \right], \quad B(\pi) = \mathbb{E} \int_t^\tau e^{-\rho(r-t)} \frac{\theta_r^2}{2} dr.$$

Then $V(k) = \sup_\pi \{A(\pi) - kB(\pi)\}$, and $0 \leq B(\pi) \leq .02C_\rho(T-t)$.

THEOREM 3—Adjustment costs and approximate optimality: *If the feasible policies and termination contract are independent of k , V is convex and nonincreasing. For $k_1 < k_2$, optimal policies satisfy $B(\pi_2) \leq B(\pi_1)$. More generally, if $V(k_i) - J_{k_i}^{\pi_i} \leq \varepsilon_i$, then*

$$B(\pi_2) - B(\pi_1) \leq \frac{\varepsilon_1 + \varepsilon_2}{k_2 - k_1}. \quad (16)$$

If measured budgets satisfy $|\widehat{B}_i - B(\pi_i)| \leq \xi_i$, their possible reversal is bounded by the right side of (16) plus $\xi_1 + \xi_2$. At a differentiability point with an attained optimizer, $V'(k) = -B(\pi_k)$.

The coefficient in (16) is sharp without further restrictions on the feasible policy set. The conclusion concerns discounted adjustment expenditure, not pointwise values of $\theta(t, u, x)$. The exact monotonicity statement is an economic theorem for the original continuous contract; the numerical experiment must still establish sufficient accuracy to quantify its magnitude.

PROPOSITION 3—Value intervals, budgets, and access to adjustment: *Suppose $L(k) \leq V(k) \leq U(k)$ and $k_- < k < k_+$. Every attained optimal adjustment budget at k belongs to*

$$\left[\max \left\{ 0, \frac{L(k) - U(k_+)}{k_+ - k} \right\}, \min \left\{ .02C_\rho(T - t), \frac{U(k_-) - L(k)}{k - k_-} \right\} \right]. \quad (17)$$

If $[L_F, U_F]$ and $[L_R, U_R]$ enclose the flexible and the restricted $\theta \equiv 0$ values, the value of access to adjustment belongs to $[L_F - U_R, U_F - L_R]$, intersected with $[0, \infty)$.

These bounds determine meaningful numerical targets. To resolve a budget difference of size d_B across prices separated by Δk , an error budget of order $\Delta k d_B$ is required, together with a budget-evaluation enclosure. An arbitrary .05 value target need not suffice. Likewise, a positive measured flexible-minus-restricted value is not a certified welfare gain unless its lower interval endpoint is positive. The utility units here are explicitly normalized lifetime utility; they have not been converted into an empirically calibrated consumption-equivalent welfare metric.

6. NUMERICAL EVIDENCE

The experiments distinguish numerical validity, achieved precision, and comparative performance. The original-payoff NDU certificate in Table I is an outward-interval bound for a continuous diffusion. The finite gain envelopes in this section are float64 calculations for their declared candidate economies. The learned-geometry experiment has a rigorous continuous verification pair but a deliberately modified running payoff. The external comparison estimates realized costs and sampling uncertainty; it does not know the optimal value. These distinctions are part of the interpretation of the results, not alternative names for the same error.

The complete new panel was executed from source commit 14ecbd48 in a clean Python 3.11 environment, with PyTorch 2.10.0, NumPy 2.3.5, SciPy 1.17.0, and mpmath 1.3.0. Numerical work used one CPU thread. The ledger retains the processor description, exact versions, source hashes, external-code hashes, checkpoint weights, rollout arrays, all seed-level records, and execution logs. Table generation reads those committed records and fails on missing cells. The local integration runs are development checks, not extra independent confirmatory seeds.

6.1. Preserving a learned policy through thresholded correction

Table II re-audits two retained policies without retraining or selecting a new policy using the reference optimum. The twelve-date seed-10 policy had a raw localized envelope of .0922420. The preceding all-positive-gain safeguard replaced 54.6602 percent of interior state-date decisions and obtained a .00544535 envelope. The threshold in (9), at target .05, replaces only .8746 percent and obtains a .00682279 envelope. At target .01, the correction share is 4.9350 percent and the envelope is .00676060. Thus most of the earlier replacements were unnecessary for meeting the declared finite-economy tolerance in this sample.

TABLE II
THRESHOLDED CORRECTION OF RETAINED NDU POLICIES

N	Target	Raw bound	Final bound	Override (%)	Audits	Seconds
6	0.05	0.042805	0.042805	0.000	1	0.472
6	0.01	0.042805	0.003278	8.796	2	0.849
12	0.05	0.092242	0.006823	0.875	2	1.641
12	0.01	0.092242	0.006761	4.935	2	1.630

Float64 finite-candidate envelopes, not continuous-model certificates. Seconds include all listed audits and envelope calculations on the current worker. Each audit still enumerates the full 125-action grid plus the proposal. Historical training time is not added across different environments.

This improvement addresses the role of correction rather than concealing it. The corrected rule differs from the raw rule, and every audit still enumerates 125 grid actions plus the neural proposal. For the twelve-date runs, two audits use 2,362,200 value-action evaluations and 2,343,600 envelope-action evaluations, each with four transition branches. The new policy is not obtained by a cheap constant-cost acceptance test. Table II includes evaluation, gain calculation, envelope propagation, and re-audit time. Historical training time is retained separately because it was measured in a different run and environment; adding it to current audit time would not be a same-machine end-to-end timing experiment.

The six-date policy already passes the .05 target without correction. At .01, it requires an 8.7956 percent correction share. These are deterministic post-review re-audits of the specified seed-10 policies, not a new population estimate of raw training success. Proposition 2 explains the condition under which correction shares vanish, but this two-policy exercise does not demonstrate a training-dose convergence curve. Neither its finite envelope nor its smaller correction share closes the continuous NDU error budget.

6.2. Separating time, state, and action approximation

Table III varies one approximation factor at a time. The time experiment fixes a 25×31 lattice and five points per action component; the state experiment fixes 48 dates and five action points; the action experiment fixes 48 dates and the 25×31 lattice. All retain the original four-sign shock rule and the straight-segment liquidation treatment. The latter two approximations are not independently refined in this panel.

The time-only sequence is not evidence of convergence. Its central value moves from -1.45149 at 24 dates to -1.60853 at 96 dates, while the exit probability rises substantially. Keeping the spatial mesh fixed makes h^2/Δ increase, rather than shrink; the experiment exposes this interaction instead of fitting an unjustified time order. Conversely, refining the state lattice at 48 dates moves the value from -1.54979 to -1.39165 . Action refinement improves

TABLE III
ONE-FACTOR NDU APPROXIMATION DIAGNOSTICS

Factor	N	State nodes	n_a	Value	Exit prob.	h^2/Δ
time	24	25×31	5	-1.451486	0.104315	0.1067
time	48	25×31	5	-1.549787	0.193358	0.2133
time	96	25×31	5	-1.608530	0.227492	0.4267
state	48	25×31	5	-1.549787	0.193358	0.2133
state	48	37×46	5	-1.444048	0.151095	0.0948
state	48	49×61	5	-1.391646	0.102389	0.0533
action	48	25×31	3	-1.606541	0.245260	0.2133
action	48	25×31	5	-1.549787	0.193358	0.2133
action	48	25×31	9	-1.536711	0.182784	0.2133

All entries use $k = 2$ and evaluate $(u, x) = (2, 1.25)$. n_a is the number of nodes per action component. Four-sign quadrature and straight-segment liquidation are unchanged; these differences do not bound continuous-model error.

the finite optimum, but the three approximation factors cannot be conflated. The operator-agreement regression is close to machine precision; agreement of two implementations of the same approximation is not an enclosure of its continuous-model bias.

A further panel uses 96 dates, a 49×61 lattice, and nine points in each action component, or 729 actions. Each adjustment-cost economy is re-solved. Table IV shows the same ordering of adjustment expenditure as Theorem 3, with an active preference margin. This is a stronger finite-action resolution than the earlier five-point panel, but still not a verified continuous welfare difference. The first-exit and quadrature terms in (15) remain unbounded by these calculations. Accordingly, neither a Richardson remainder nor a continuous policy-convergence rate is inferred from this table.

TABLE IV
FINER FINITE-ACTION PREFERENCE-ADJUSTMENT COMPARISON

k	Value	Budget B	Exit prob.	θ_0	p_0
0.5	-1.416240	0.016811	0.162725	0.2000	0.3125
2	-1.437379	0.011583	0.162286	0.2000	0.3125
8	-1.468043	0.001888	0.158876	0.1000	0.3125

$N = 96$, a 49×61 state lattice, and nine nodes per action component (729 actions). Central consumption is .8 in these rows. Values and budgets belong to the declared numerical liquidation economy, without a continuous error enclosure.

6.3. A learned actor on the full NDU control and exit geometry

To test whether continuous verification transfers from an analytic example to an actually trained control, retain (2), the three-dimensional action box, correlated shocks, state domain, and settlement exactly. Change only the running payoff to

$$\tilde{\ell}(t, s, a) = \ell_2(s, a) - \sup_{b \in A} R_b g(t, s). \quad (18)$$

The subtracted term is independent of the chosen action, but depends on state and time. Thus it generally changes optimal behavior relative to the original-payoff economy; it is not an

innocuous preference normalization. In the modified problem $\sup_a \tilde{R}_a g = 0$ and the explicit feasible maximizer is $(.8, -.02(u - 2), .8)$. Itô verification proves that the modified value is g with the original stopping contract.

We train a four-parameter actor with constant bounded sigmoid consumption and portfolio outputs and an affine-tanh adjustment output. The actor maximizes the Hamiltonian of the known witness, with no optimal-action labels. Six declared seeds each receive 4,000 Adam steps at learning rate .02 and 256 random states per step. This is a controlled verification experiment, not a deep-network capacity or high-dimensional scaling claim. The witness is known rather than learned; the actor and all of its achieved errors are learned.

For the frozen weights, the control gap is bounded over the entire state domain by outward interval arithmetic. The consumption gain is bounded using

$$\int_c^{.8} z^{-u} dz - .1(.8 - c)/x \leq (.8 - c)\{c^{-2.8} - .05\},$$

the portfolio gain is the exact quadratic difference, and the adjustment gain equals $(\theta + .02(u - 2))^2$ at $k = 2$. A 2,048-cell interval partition in u encloses the latter. The full regret is bounded by $C_\rho(1)$ times the summed action-gap enclosure. The time discretization of a rollout and a numerical first-exit probability do not enter this certificate, because its proof is for the original continuous generator.

TABLE V

ALL-STATE CONTINUOUS CERTIFICATE FOR TRAINED NDU-GEOMETRY ACTORS

Seed	Regret bound	c error	θ error	p error	Train (s)
300	0.00081494	0.0004555	0.0000318	0.0007034	2.774
301	0.00103566	0.0005788	0.0000344	0.0007415	2.749
302	0.00081053	0.0004532	0.0000308	0.0005803	2.778
303	0.00112262	0.0006277	0.0000397	0.0005607	2.761
304	0.00112212	0.0006273	0.0000635	0.0006387	2.794
305	0.00099308	0.0005551	0.0000334	0.0007077	2.780

All-state upper bounds use outward interval arithmetic and the same state, action, covariance, and first-exit geometry as the economic model. Only the running payoff is manufactured. Four actor parameters are trained for 4,000 steps without optimal-action labels; the critic witness is known.

Every seed meets the .01 modified-problem target, with continuous regret bounds between .00081053 and .00112263. Table V separately reports componentwise control errors rather than treating value accuracy as policy sup-norm accuracy. This is the desired transfer test with a learned actor and the original first-exit/control geometry. It does not identify the original NDU value, and does not tighten Table I merely by sharing the state equation.

6.4. Coupled nonconvex control and an external author implementation

The new comparison does not use the Cole–Hopf quadratic-control model or regress to an analytic greedy target. Starting from $X_0 = 0$ in $\mathbb{R}^d$, minimize

$$\mathbb{E} \left[\int_0^1 f(a_t) dt + q(X_1) \right], \quad dX_t = 2a_t dt + \sqrt{2}(.4)dW_t, \quad a_t \in [-1, 1]^d, \quad (19)$$

$$f(a) = d^{-1} \sum_{i=1}^d (a_i^2 + .1 \sin^2 a_i) + .3 \sin \left(\sum_i a_i \right), \quad (20)$$

$$q(x) = d^{-1} \sum_i (x_i - .5)^2 + .2 \cos \left(d^{-1/2} \sum_i x_i \right). \quad (21)$$

The sine term couples all control components. For example, at an interior vector whose sum is $\pi/2$, the Hessian of the running cost has a negative direction along the all-ones vector for both tested dimensions. Hence the problem is genuinely nonconvex in the action. No closed-form maximizer or optimal-control labels are supplied to either method, and the quadratic exponential transformation does not eliminate this Hamiltonian.

The external comparator is the unmodified `SOCMartNet.train` routine in the authors' repository `sx-fang/MartNet`, pinned at commit `991ea8dd`. Its complete module hashes are in the execution manifest. The integration consists of model and network adapters and a scheduler that stops after a completed update at the time budget. The authors' solver is not rewritten as a bespoke PINN comparator. The pinned implementation uses a Hessian-free Hamiltonian interface; we therefore use constant diffusion in this comparison. This is a restriction of the implemented interface used here, not an assertion that the published SOC-MartNet framework excludes volatility control.

Both methods receive identical initial actor and critic weights within each seed. Both use two tanh hidden layers of width 48, bounded actor outputs, a terminal-exact critic, 16 training time intervals, and one CPU thread. NBO uses three evaluation updates per improvement update, 256 sampled states, four Gaussian successors, and Adam with learning rate `.003`. SOC-MartNet retains its additional test network, RMSprop updates, and the declared martingale and Hamiltonian losses. Its actor/value learning rate is `.003/ $\sqrt{d}$` , the test-network learning rate is `.01`, and the remaining adapter settings are fixed in the protocol. The test network has 600 outputs; its parameters and all setup costs are reported rather than excluded from the method description.

The primary budget is ten seconds of training per method, allowing the final update to complete. Setup time is outside this primary budget and is reported separately. This matters: the mean setup-plus-training time is approximately 10.69 seconds for NBO and 10.02 seconds for SOC-MartNet on this worker. Therefore the experiment is not described as an exactly matched ten-second end-to-end race. NBO is evaluated first in each process, so one-time framework initialization is included in its measured setup cost. Table VII makes the distinction visible.

TABLE VI
PAIRED NONCONVEX-CONTROL COMPARISON AT A FIXED TRAINING BUDGET

d	NBO cost	SOC cost	Difference	Paired 95% interval
8	0.59567	2.31352	-1.71785	$[-2.13823, -1.29747]$
16	0.48067	2.72799	-2.24732	$[-2.51288, -1.98176]$

Difference is NBO minus SOC-MartNet realized cost at 128 audit steps; negative favors NBO. Each dimension has six independent training/audit seeds, 4,096 common-random-number paths per pair, and ten seconds of training per method. Student intervals describe across-seed uncertainty, not optimality loss. Maximum paired path standard errors are 0.01098, 0.00900, respectively.

At dimension eight, the paired mean NBO-minus-SOC cost is -1.71785 , with a six-seed Student interval $[-2.13823, -1.29747]$. At dimension sixteen it is -2.24732 , with interval $[-2.51288, -1.98176]$. Thus the new evidence favors NBO under the specified adapter and

TABLE VII
MEASURED TRAINING AND SETUP RESOURCES

d	Method	Train (s)	Setup + train (s)	Outer updates
8	NBO	10.003	10.692	1536.7
8	SOC-MartNet	10.012	10.015	465.3
16	NBO	10.004	10.689	1477.3
16	SOC-MartNet	10.012	10.014	458.0

Single-thread CPU means over six seeds. Identical actor/critic parameter counts within each dimension: 3,224/2,881 at $d = 8$, and 4,000/3,265 at $d = 16$. SOC-MartNet also trains its 600-output test network. Paired policy audit means are approximately .379 and .440 seconds. Outer-update definitions differ across algorithms and are not equal units of work.

short training budget. This is a demonstrated benefit in a coupled nonconvex problem, in contrast to the retained R5 quadratic-control comparison. It is not a theorem of computational dominance, a replication of the published SOC-MartNet hardware or learning curves, or a statement that either computed policy is near optimal.

6.5. Uncertainty, mesh diagnostics, and retained failures

Each paired policy audit uses 4,096 paths with common Brownian increments for the two policies and an independent seed for each training seed. The primary outcome uses 128 time steps. A nested 64-step audit aggregates adjacent increments, so its difference from the finer audit is a matched time-mesh diagnostic. Table VI reports the maximum paired path standard error separately from the across-seed interval. The interval is based on six independent training/audit seeds, with a Student approximation rather than a distribution-free coverage claim. Paths are not counted as independent training runs.

The average 128-minus-64 cost changes are about $-.00149$ and $-.00243$ for NBO and SOC-MartNet at dimension eight, and $-.00176$ and $.00951$ at dimension sixteen. These changes are much smaller than the observed paired cost differences, but they are not certified bounds on the remaining time-discretization error. A matched-accuracy computational frontier and longer-budget learning curves would answer additional questions not identified by the present fixed-budget design.

All prior outcomes remain part of the evidence. The R5 ten-seed baseline had ten raw finite-target passes, whereas its three twelve-date raw policies failed. Its recursive raw policies missed their own target, and its eight/sixteen/thirty-two-dimensional quadratic-control NBO means were worse than those of the in-house PINN-PI comparator. Those results are neither relabeled as successes nor removed after observing the new comparison. The R4 Merton, nonlinear NDU, cost, boundary, recursive, temporal-self, game, and coupled linear-quadratic calculations, including failed pilots and incomplete runs, are reproduced in the preservation supplement. The new external experiment changes the tested model and comparator; it does not retrospectively change those historical results.

7. RELATION TO NEIGHBORING NUMERICAL METHODS

Table VIII separates the relevant methodological axes. Its entries summarize the cited analyses and the implemented comparison; they are not a ranking by paper title. In particular, an interior PDE error theorem with unknown outer data, a stopped-control regret theorem with a contractual boundary, and a finite candidate-action gain envelope address different mathematical objects.

For an analyst choosing a method, the distinction between state dimension and action dimension is crucial. Neural state representations avoid a tensor state grid during the high-dimensional

TABLE VIII
RELATED METHODS AND THE OBJECTS COMPARED

Method	Principal mechanism or guarantee	Relation to the present evidence
Exploratory PINN policy iteration	Soft policy updates for entropy-regularized HJB equations; iteration, policy-network, and PDE residual errors	Neighboring policy-evaluation/improvement analysis; not the external implementation tested here
Interior physics-informed policy iteration	Stationary interior error control with residual and outer-information contributions under stated regularity conditions	Clarifies what interior collocation can control without supplying the stopped NDU boundary data
PI-DeepONet	Policy iteration and reusable operator approximation across terminal data	Representation reuse is a different performance target from one trained policy at a fixed budget
SOC-MartNet	Martingale training of value and control without an explicit Hamiltonian infimum	Pinned author routine tested on coupled constrained nonconvex controls in $d = 8, 16$
NBO in this paper	Separated evaluation, improvement, and independent gain or witness verification	Continuous stopped-model theorem, explicit economic error calibration, finite correction costs, and paired author-code experiment

Sources: Kim, Cho, Kim, and Kim (2026), Kim, Kim, Kim, and Cho (2026), Lee and Kim (2025), Cai, Fang, and Zhou (2025). The new external benchmark uses constant diffusion and a ten-second training budget. Its scope does not establish a general ranking for recursive utilities, games, or higher dimensions.

training experiment, while the finite NDU safeguard still has tensor action cost. The exact component selection in Proposition 1 is specific to that economic Hamiltonian. The coupled sine-control problem deliberately removes that separability. Likewise, none of the numerical comparisons establishes that a small sampled training loss satisfies the uniform hypotheses of the continuous certificate. The contribution is a verifiable separation of these issues, with one favorable external comparison and one original-economy precision test whose conservative outcome remains visible.

8. RECURSIVE UTILITY AND STRATEGIC DECISIONS

8.1. Recursive utility

Let $a = 1 - 1/\psi$, $b = 1 - \gamma$, and $bV > 0$. The retained Epstein–Zin normalization is

$$f(c, V) = \frac{\rho}{a} c^a (bV)^{1-a/b} - \frac{\rho b}{a} V, \quad (22)$$

with the appropriate logarithmic limit when $a = 0$; see Epstein and Zin (1989), Duffie and Epstein (1992). Evaluation is $V_t + \mathcal{L}^{c,p}V + f(c, V) = 0$, not an additive-reward equation with the preference parameters renamed. The transform $V = \exp(W)/b$ enforces the sign domain. A compact positive margin for bV , bounds on consumption, and a monotone implicit step are additionally needed for a finite Lipschitz bound on the evaluation operator.

The gain argument extends beyond affine evaluation. If the action operators $\mathcal{T}_n^a$ are monotone and uniformly Lipschitz with constants L_n on a verified continuation-value interval, then (8) holds with δ^{j-n} replaced by $\prod_{m=n}^{j-1} L_m$, and the terminal factor by $\prod_{m=n}^{N-1} L_m$. Appendix D

proves this statement. A sign-enforcing architecture alone does not verify the required compact interval or monotonicity.

The supplementary record retains the stationary homothetic anchors $p = .3$ and $c/x = .0455$ at the declared parameters, the executed implicit recursive reference, and the R5 trained recursive runs. In the latter, raw losses around .0425 and .0502 against a .01 target were corrected using a 961-action reference. Those corrected results establish finite-model performance of the corrected rule; they do not establish small correction cost or a scaled neural recursive-portfolio solution. These experiments and their limitations are preserved rather than replaced by an additive-control result.

8.2. *Sophisticated temporal selves*

For a sophisticated acting self with present-bias parameter β and ordinary discount δ , the improvement and evaluation equations differ:

$$c_n(s) \in \arg \max_c \{u(c)\Delta + \beta\delta\mathbb{E}[V_{n+1}(F(s, c, \varepsilon))]\}, \quad (23)$$

$$V_n(s) = u(c_n(s))\Delta + \delta\mathbb{E}[V_{n+1}(F(s, c_n(s), \varepsilon))]. \quad (24)$$

Only improvement carries the extra β . With two consumption dates, terminal log wealth, zero interest, and $\delta = 1$, the sophisticated first-date share is $1/(1 + 2\beta)$, hence .4166667 at $\beta = .7$. Replacing ordinary evaluation by geometric- β evaluation gives .4566210, a different behavioral model. The retained derivation, executed residual check, and feasible-deviation calculation remain part of the supplement. Equilibrium verification concerns each acting self's feasible deviation with future selves fixed, not a fictitious commitment planner who can change all future selves' preferences.

8.3. *Dynamic games*

For player i , fix the rivals' feedbacks and solve both the induced policy-evaluation problem and a unilateral best-response problem over all subsequent dates and states. Applying the finite or continuous gain certificate to this best-response problem bounds player i 's exploitability. A Nash certificate requires this test for every player. Differentiating joint profits, allowing gradients through rivals, or checking only realized-path deviations is a different operation.

The retained static Cournot and three-date stochastic capacity-game records test these distinctions. The dynamic game has two firms, capacity investment, stochastic demand, and multiple equilibrium selections. The R5 neural game adds actual actor training, but its eighteen states and analytic best-response verification remain small-scale. Neither experiment establishes general equilibrium scalability or a competitive limit. The operator formulation, nonzero present bias, recursive aggregator, and strategic state transitions are retained as substantive extensions; larger certified experiments on these objects remain a separate quantitative requirement, not a consequence of the new additive nonconvex benchmark.

9. COMPUTATION WITH STOCHASTIC TRACES

If A is symmetric and z_k are independent standard Gaussian probes, put $\bar{q}_K = K^{-1} \sum_k z'_k A z_k$. Then

$$\mathbb{E}(c - \bar{q}_K/2)^2 = (c - \text{tr } A/2)^2 + \frac{\|A\|_F^2}{2K}. \quad (25)$$

Averaging separately squared one-probe residuals retains a variance penalty that does not decrease with K . Two independent probe means give an unbiased cross-product estimator of the squared exact residual, but not a nonnegative sample objective or a uniform residual enclosure. The proof, variance accounting, and the retained Monte Carlo regression are given in the supplement and Appendix E.

For a diffusion loading with m columns, an exact contraction can use m directional second derivatives; a randomized contraction uses K probe operations and the loading operations. Network size, parameter differentiation, collocation coverage, optimization iterations, and the accuracy-dependent probe budget all enter total cost. Avoiding a dense Hessian is useful, but does not imply a universal dimensional crossover or eliminate the cost of verification. The retained coupled linear–quadratic tests with off-diagonal drift, rewards, and shocks are useful regression checks, while their known quadratic value family and faster Riccati comparator preclude a general performance conclusion.

10. CONCLUSION

NBO makes a policy-evaluation approximation, an improvement rule, and a verification calculation separately inspectable. In a stopped preference–portfolio economy, this separation leads to a complete constrained Hamiltonian audit, one-sided continuous value witnesses, and quantitative conditions for adjustment-cost comparisons at finite numerical accuracy. The new learned-geometry certificate and paired author-code comparison address different questions: transfer of a verification method and realized performance on coupled nonconvex controls. Neither is used to erase the conservative original-payoff certificate or the substantial cost of a finite safeguard.

The economic scope continues to include recursive utility, sophisticated temporal selves, and strategic interaction, with their distinct evaluation and equilibrium conditions. The remaining numerical task is concrete: construct sufficiently narrow original-model witnesses, or enclose every term of the declared semigroup error budget, so that the economic effect exceeds the verified approximation error. Complete current source, retained derivations, executed records, and a source-bound build make this requirement available for the next independent review.

APPENDIX A: PROOFS OF THE CONTINUOUS VERIFICATION RESULTS

A.1. *Proof of Theorem 2*

For a smooth witness w and a fixed admissible policy a , localized discounted Itô calculus gives

$$\mathbb{E} \left[e^{-\rho(\sigma-t)} w(\sigma, S_\sigma) + \int_t^\sigma e^{-\rho(r-t)} \ell(S_r, a_r) dr \right] = w(t, s) + \mathbb{E} \int_t^\sigma e^{-\rho(r-t)} R_{a_r} w(r, S_r) dr, \quad (26)$$

where σ first stops on a compact interior localization or at the economic stopping time. Localization also bounds the stochastic integral. The hypotheses on the witness and uniform integrability permit passage to τ ; no regularity of the optimal value is invoked.

For $w = U$, $R_a U \leq \epsilon_U$ for every admissible action. The terminal inequality $g \leq U + b_U$ and $e^{-\rho(\tau-t)} \leq 1$ yield

$$J^a(t, s) \leq U(t, s) + b_U + \epsilon_U \mathbb{E} \int_t^\tau e^{-\rho(r-t)} dr \leq U(t, s) + b_U + C_\rho(T - t) \epsilon_U.$$

Take the supremum over policies; attainment is unnecessary. For $w = L$, use the specified policy π , the inequalities $R_\pi L \geq -\epsilon_L$ and $g \geq L - b_L$, and the same integration identity to obtain the lower bound. Subtraction proves the regret bound, and nonnegativity follows from the definition of the supremum.

For multiple time slabs, apply (26) on each slab and sum. Forward-time jumps contribute $U(t^+, S_t) - U(t^-, S_t)$ to the upper-witness identity. They are nonpositive under Assumption 1, so the upper inequality is preserved. Lower-witness jumps have the opposite sign. This explains why a piecewise-time approximation may not silently omit its interface errors. The actual policy can be only piecewise continuous in time because Itô's formula is applied to the witnesses, not to a derivative of the policy. Q.E.D.

A.2. Spatial residual potentials

Suppose the upper residual satisfies $R_a U \leq q_U$ and a nonnegative potential obeys

$$z_{U,t} + \mathcal{L}^a z_U - \rho z_U \leq -q_U \quad \text{for every } a.$$

Then $R_a(U + z_U) \leq 0$ by the affine dependence of R_a on the witness. Its boundary inequality is preserved by nonnegativity of z_U . Similarly, if $R_\pi L \geq -q_L$ and $z_{L,t} + \mathcal{L}^\pi z_L - \rho z_L \leq -q_L$, then $R_\pi(L - z_L) \geq 0$. Theorem 2 applies to $U + z_U$ and $L - z_L$. A potential solving an occupation problem exactly would weight residuals by reachable discounted occupation; an approximate potential is usable only after the displayed differential inequality is enclosed. This construction does not turn an empirical occupancy histogram into a theorem.

A.3. Verification of the explicit original-economy witnesses

Since $\partial_t C_\rho(1-t) = -e^{-\rho(1-t)}$ and $\rho C_\rho(1-t) = 1 - e^{-\rho(1-t)}$,

$$(\partial_t - \rho)[-aC_\rho(1-t)] = a$$

for a constant a . Hence $\sup_a R_a G \leq -\alpha$ implies $\sup_a R_a U \leq 0$, and $R_\pi g \geq -\beta$ implies $R_\pi L \geq 0$. On the stopped boundary,

$$U - g = 8(1-t) - \alpha C_\rho(1-t) \geq (8-\alpha)(1-t) \geq 0, \quad L - g = -\beta C_\rho(1-t) \leq 0.$$

Both witnesses equal G at maturity. They are smooth and bounded on the closed state rectangle because $x \geq .5$. The bilinear policy is globally Lipschitz on this compact rectangle, component-wise feasible, and piecewise constant in its time index. The stopped coefficients are bounded and have the required state regularity, so the discounted stochastic integrals are integrable. The sign enclosure therefore proves a statement about the original continuously stopped policy, including policies for which the optimal boundary regularity conditions of a different PDE theorem fail.

The interval calculation bounds all a in the upper witness through the exact maximizing rule for G . It bounds the lower witness on every time-state cell by the convex range of the four policy corners, enlarged to a Cartesian box; enlarging the action range can only make the lower bound more conservative. Interval dependency can likewise enlarge the bound, never certify a narrower true range. No assertion of floating-point simulation conformance is needed: the mathematical policy is defined by the stored constants and exact interpolation weights. A separate deployment-error allowance would be required for another implementation.

A.4. Proof of Proposition 1

The action-dependent part of $R_a v$ is the sum of

$$\frac{c^{1-u}}{1-u} - v_x c, \quad v_u \theta - \frac{k}{2} \theta^2, \quad h_1 p + \frac{h_2}{2} p^2.$$

The mixed state derivative changes h_1 but introduces no product between control components. The consumption objective has derivative $c^{-u} - v_x$ and strictly negative second derivative $-u c^{-u-1}$. If $v_x \leq 0$, its derivative is positive everywhere. Otherwise its unique unconstrained stationary point is $v_x^{-1/u}$, and projection gives the constrained optimum. The adjustment objective is a strictly concave quadratic because $k > 0$. The portfolio objective is strictly concave when $h_2 < 0$, linear when $h_2 = 0$, and convex when $h_2 > 0$; the stated vertex or endpoint rule is complete in all cases. Compact product feasibility and additivity make these separate maximizers a global joint maximizer. Q.E.D.

A.5. A learned-geometry regret bound

For (18), let $H_g^*(t, s) = \sup_a R_a g(t, s)$ for the original running payoff. The modified residual is $\tilde{R}_a g = R_a g - H_g^* \leq 0$, with equality for the feasible explicit optimizer. Applying (26) first to any policy and then to this optimizer proves $\tilde{V} = g$. For the learned policy,

$$g(t, s) - \tilde{J}^\pi(t, s) = \mathbb{E}_{t,s}^\pi \int_t^\tau e^{-\rho(r-t)} \{H_g^* - R_\pi g\}(r, S_r) dr.$$

The action-independent terms cancel. At $k = 2$ the adjustment term completes the square to $(\theta + .02(u-2))^2$. The other two action differences are exactly those bounded in Section 6.3. Their all-state interval upper bound times $C_\rho(1-t)$ gives the recorded continuous regret certificate. Policy sup-norm errors are separate interval consequences, not inferred by reversing a regret inequality. Q.E.D.

APPENDIX B: POLICY ITERATION AND FINITE GAIN ACCOUNTING

B.1. The exact separated-operator limit

Let $V_j = \mathcal{E}\pi_j$ and $\pi_{j+1} = \mathcal{I}(V_j)$. The selector inequality implies $R_{\pi_{j+1}} V_j \geq 0$, so exact comparison and verification yield $V_{j+1} \geq V_j$. Precompactness of the evaluation image bounds the sequence. Its pointwise increasing limit is denoted $\bar{V}$.

Every subsequence has a further subsequence converging in the declared norm of the function, time derivative, gradient, and Hessian. The function component of any such limit must equal $\bar{V}$, by pointwise monotonicity. Its derivative components are the derivatives of that same function limit. Thus the whole value sequence converges in that strong norm: otherwise a subsequence separated from the unique possible strong cluster point would contradict precompactness.

Along a policy subsequence $\pi_{j_m} \rightarrow p$, continuity of evaluation gives $\mathcal{E}p = \bar{V}$. Continuity of the selector at this cluster point gives $\pi_{j_m+1} \rightarrow \mathcal{I}(\bar{V})$, and continuity of evaluation gives $\mathcal{E}(\mathcal{I}(\bar{V})) = \bar{V}$. The evaluation equation of the improved limiting policy therefore reads

$$R_{\mathcal{I}(\bar{V})} \bar{V} = 0.$$

The defining maximizing property of that selector implies $\sup_a R_a \bar{V} = 0$. Boundary comparison identifies $\bar{V} = V^*$. The proof establishes equality of limiting values, not equality of consecutive

limiting policies. Compactness, selector closure, and strong evaluation continuity are hypotheses, not properties of an arbitrary neural architecture or Adam trajectory. *Q.E.D.*

B.2. Proof of Theorem 1

Positivity and substochasticity give $\|\mathcal{T}_n^a v - \mathcal{T}_n^a w\|_\infty \leq \delta \|v - w\|_\infty$. The maximized operator has the same Lipschitz constant. Since the candidate set includes the enacted action,

$$\|v_n - \max_a \mathcal{T}_n^a v_{n+1}\|_\infty \leq e_n + \eta_n.$$

Comparing this approximate equation with optimal backward induction yields

$$\|v_n - V_{h,n}^*\|_\infty \leq \delta^{N-n} b + \sum_{j=n}^{N-1} \delta^{j-n} (e_j + \eta_j).$$

Comparing instead with policy evaluation gives the same inequality with only e_j in the sum. Add the two bounds to prove (8).

For the localized bound let $d_n = V_{h,n}^* - V_{h,n}^\pi$. Exact evaluation implies $g_n \geq 0$. For every state, the elementary maximum-difference inequality gives

$$d_n \leq g_n + \delta \max_a P_n^a d_{n+1}.$$

Starting with $d_N = 0$, induction and positivity give $d_n \leq B_n$. The reference optimal value is unnecessary in this recursion: only policy evaluation, transition matrices, and feasible gains are used. *Q.E.D.*

B.3. Threshold correction and arithmetic scope

If every recomputed gain is at most γ , Theorem 1 gives a regret bound at most $\gamma \sum_n \delta^n$. Thus (9) is sufficient with margin when this uniform condition holds. Correcting a subset of decisions does not by itself guarantee that condition, since the re-evaluated continuation changes. The implementation therefore re-evaluates and re-audits rather than declaring success after a fixed number of replacements. Its stopping certificate is the actual envelope of the corrected policy.

For Proposition 2, let I_m indicate $g_m > \gamma$. Then $\gamma I_m \leq g_m$, so $M^{-1} \sum_m I_m \leq (M\gamma)^{-1} \sum_m g_m$. The fixed- and shrinking-threshold implications follow directly. This statement measures the correction share under the audit measure; it does not imply that all reachable-state losses have small occupation-weighted averages or that a full audit is computationally free. *Q.E.D.*

The finite envelopes reported in the numerical tables are float64 quantities with independent operator-agreement and regression tests. They are not outward-rounded enclosures of every finite arithmetic operation. To make a machine-rigorous finite certificate one must add roundoff and inner-solve enclosures to e_n and η_n , or evaluate their analogues with validated arithmetic. The continuous witness and learned-geometry certificates do use outward interval arithmetic, and are identified separately for this reason.

B.4. Proof of the continuous semigroup error decomposition

With settlement fixed inside each step, the exact surviving-continuation map has Lipschitz constant at most δ : survival probabilities are at most one and continuation is received at the

full step end. Taking a supremum over all admissible within-step controls preserves this bound. Write $E_n^* = \|V_n - w_n^*\|_\infty$ and $E_n^\pi = \|J_n^\pi - w_n^\pi\|_\infty$. Then

$$E_n^* \leq \kappa_n^* + \delta E_{n+1}^*, \quad E_n^\pi \leq \kappa_n^\pi + \delta E_{n+1}^\pi.$$

Backward iteration and the positive norm-one lift of the finite gap prove (15). If $\mathcal{B}^{(0)}, \dots, \mathcal{B}^{(m)}$ form an intermediate operator chain from the exact semigroup to the lifted numerical operator, the triangle inequality bounds its discrepancy on a specified continuation w by $\sum_i \|(\mathcal{B}^{(i)} - \mathcal{B}^{(i+1)})w\|_\infty$. Every term must have a verified bound; changing the tested continuation between adjacent differences without controlling that change invalidates the telescoping argument. *Q.E.D.*

APPENDIX C: PROOFS FOR ECONOMIC COMPARISONS

For a fixed policy, $A(\pi) - kB(\pi)$ is affine and nonincreasing in k . The supremum is therefore convex and nonincreasing. The budget is bounded by $.02C_\rho(T - t)$ because $|\theta| \leq .2$. If π attains the value at k , for every other price k' ,

$$V(k') \geq A(\pi) - k'B(\pi) = V(k) - (k' - k)B(\pi).$$

Thus $-B(\pi)$ is a subgradient of V at k . At a differentiability point every attained optimizer has budget $-V'(k)$.

C.1. Proof of Theorem 3

Approximate optimality and feasibility of the rival candidate policy give

$$\begin{aligned} A(\pi_1) - k_1 B(\pi_1) &\geq A(\pi_2) - k_1 B(\pi_2) - \varepsilon_1, \\ A(\pi_2) - k_2 B(\pi_2) &\geq A(\pi_1) - k_2 B(\pi_1) - \varepsilon_2. \end{aligned}$$

Adding and collecting terms yields $(k_2 - k_1)\{B(\pi_2) - B(\pi_1)\} \leq \varepsilon_1 + \varepsilon_2$. The exact monotonicity statement is the zero-error case. Adding the two budget measurement discrepancies yields the stated observable-error inequality. Feasible-set independence of k is essential: otherwise a policy used in the crossed comparison might not be available.

For sharpness in the class of affine policy-value problems, take two policies with $B_1 = 0$, $B_2 = (\varepsilon_1 + \varepsilon_2)/(k_2 - k_1)$, $A_1 = 0$, and $A_2 = k_1 B_2 + \varepsilon_1$. At k_1 , policy 1 is exactly ε_1 suboptimal; at k_2 , policy 2 is exactly ε_2 suboptimal. Equality holds in the budget-reversal bound. Choose errors small enough to respect any prescribed positive upper bound on B_2 . This construction proves that the coefficient cannot be improved without more structure; it is not a claim that these two artificial policies have been realized in the particular NDU dynamics. *Q.E.D.*

C.2. Proof of Proposition 3

Use the subgradient inequality with the optimizer at k and prices k_+ and k_- . Rearrangement gives

$$\frac{V(k) - V(k_+)}{k_+ - k} \leq B(\pi_k) \leq \frac{V(k_-) - V(k)}{k - k_-}.$$

Replacing each numerator by its appropriate outer value enclosure gives (17), and intersecting with the primitive budget range is valid. Inclusion of restricted policies in the flexible set gives $V_F - V_R \geq 0$. The interval subtraction rule gives the other endpoints. No optimizer differentiability, common selected policy, or pointwise action ordering is required. *Q.E.D.*

APPENDIX D: RECURSIVE AND EQUILIBRIUM GAIN BOUNDS

Suppose a verified interval of continuation values is invariant under all action operators, each operator is monotone on it, and each has Lipschitz constant at most $L_n \geq 0$. The maximized operator is also L_n -Lipschitz. Repeating the two approximate-equation comparisons in the finite proof gives

$$\begin{aligned}\|V_n^* - v_n\|_\infty &\leq e_n + \eta_n + L_n \|V_{n+1}^* - v_{n+1}\|_\infty, \\ \|V_n^\pi - v_n\|_\infty &\leq e_n + L_n \|V_{n+1}^\pi - v_{n+1}\|_\infty.\end{aligned}$$

Iteration proves the product-weighted error formula stated in the text. These hypotheses must be verified for the actual recursive discretization. In particular, the sign condition $bV > 0$ does not exclude arbitrarily large derivatives of the driver close to zero and does not supply a uniform L_n .

For a game, fix rivals' strategies and regard player i 's unilateral deviations over all future dates as a control problem with its own payoff and transition law. The preceding certificate then bounds $V_i^{\text{BR}(\pi_{-i})} - V_i^\pi$. Requiring this bound for every player gives an approximate Nash condition at every initial state included in the norm. Checking only a single-date action deviation or only states visited in one rollout does not replace the all-future-date best-response calculation.

For sophisticated temporal selves the acting self's criterion uses the β -weighted continuation in (24), but ordinary continuation evaluation uses δ . A unilateral deviation certificate must use the acting self's criterion, with future selves held fixed. Substituting a geometrically β -discounted evaluation equation changes that equilibrium problem, so its small residual would certify the wrong object. The preserved supplement supplies the full two-date algebra and the executed finite-model conventions. *Q.E.D.*

APPENDIX E: TRACE ESTIMATION AND LIMITATIONS OF LOCAL TRAINING

For a symmetric matrix A and $z \sim N(0, I)$, diagonalize A orthogonally. Rotational invariance and independence of the coordinates imply $\mathbb{E} z' A z = \text{tr } A$ and $\text{Var}(z' A z) = 2\|A\|_F^2$. The mean of K independent probes has variance $2\|A\|_F^2/K$. Bias-variance decomposition of $(c - \bar{q}_K/2)^2$ proves (25).

Averaging the K quantities $(c - z'_k A z_k/2)^2$ leaves the expectation equal to $(c - \text{tr } A/2)^2 + \|A\|_F^2/2$. With two independent probe means $\bar{q}$ and $\bar{q}'$, independence instead gives

$$\mathbb{E}[(c - \bar{q}/2)(c - \bar{q}'/2)] = (c - \text{tr } A/2)^2.$$

The latter sample objective may be negative and may have high variance. None of these expectation identities is a uniform residual enclosure over state space. *Q.E.D.*

A local actor stationary point also cannot substitute for a global feasible action maximum. For $H(a) = -(a^2 - 1)^2 + .2a$ on $[-2, 2]$, the local maximum near $-.973994$ and the global maximum near 1.024120 are both isolated stationary points, with objective gap about $.399875$. Exact critic evaluation and isolated actor stationary points do not eliminate this difference. Concavity in the action would give the global supporting-hyperplane gap, but concavity in an action argument does not imply concavity in shared network weights.

The retained viscosity counterexample has the same logical role. A small integrated strong residual for a smooth approximation of $|x| - 1$ does not select the viscosity solution $1 - |x|$ of the appropriate eikonal boundary problem. Missing viscosity inequalities cannot be replaced by a loss plot. The previous derivations, exact boundary specifications, and regression checks for these examples are reproduced unchanged in the preservation supplement.

Likewise, a two-timescale stochastic-approximation theorem requires bounded iterates, appropriate regularity, stable fast evaluation dynamics, martingale-noise control, and step-size conditions. Its generic conclusion is an invariant-set statement for the limiting slow dynamics. An optimal-policy conclusion needs an additional argument ruling out nonoptimal attractors. The current regret theorems do not rely on asserting that every observed gradient trajectory has such an optimal limit.

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